(2001), who report that 50% of HFs managers disappear within thirty months and fewer than 5% last more than ten years. Gregoriou (2002) also finds the median survival time of HFs to be around five years. Panel B of Figure 17.2 shows that the probability of a HF failing rises steeply from the fund's inception and peaks at around three years. HFR reports that 60% of the HFs that underperformed the stock market during the financial crisis in 2008 shut down.[27]

3.3. PERFORMANCE

Hedge Fund Performance Has Declined over Time

Many studies have looked at whether HFs outperform on a risk-adjusted basis.[28] The conclusions are mixed, but I believe HFs probably did add value in the 1980s, 1990s, and even possibly in the early years of 2000. But it is unlikely that HFs add value today, on average, after adjusting for risk.

In Figure 17.3, I take the hedge fund index constructed by HFR. Panel A plots the alpha of the HF index from January 1995 to December 2012, which is computed by rolling five-year regressions on the S&P 500 at the monthly frequency. (See chapter 10 for details on alpha.) During the mid-1990s, HFs returned more than 10% on a risk-adjusted basis. This dipped to 5% in 1998—the year Russia defaulted and LTCM failed. HF alphas shot back up to around 10% from 2000–2004. Since then, the value-added by HFs has been on a downward trajectory. In the post-2008 sample, HF alphas have been returning only 1% to 2% after adjusting for market risk. In the last two years of the sample, 2011 and 2012, HF alphas were negative.

Panel B of Figure 17.3 shows that while HF alphas have been falling, the correlation of HF returns with the S&P 500 has been rising. During the late 1990s, the correlation was only 50%, meaning that HFs could add large diversification benefits to an investor's portfolio (see chapter 3). Since then, the correlation has gradually increased and in 2012 and 2013 was above 85%. With such high correlations to the S&P 500, combined with negative alphas, HFs are certainly not a separate asset class; today they are probably subtracting value from investors on a risk-adjusted basis.

Larger Hedge Funds Do Worse

Dichev and Yu (2011) compute dollar-weighted returns for the whole HF industry and uncover a sorry picture of HF performance. From 1980 to 2007, the HF industry underperformed the S&P 500, with returns of 9.7% compared to 13.1% for the stock index. Including the following year, 2008, takes things from bad to

[27] Reported in Chung, J., "A Volatile Investor Buys Into a Softer Approach," *Wall Street Journal*, March 20, 2013.

[28] In addition to the references in the text, the "no, HFs don't add value" camp also includes Ackerman, McEnally, and Ravenscraft (1999) and Hasanhodzic and Lo (2007).